2026 04 16
발표 자료

광운대학교 로봇학과
FAIR Lab

김한서

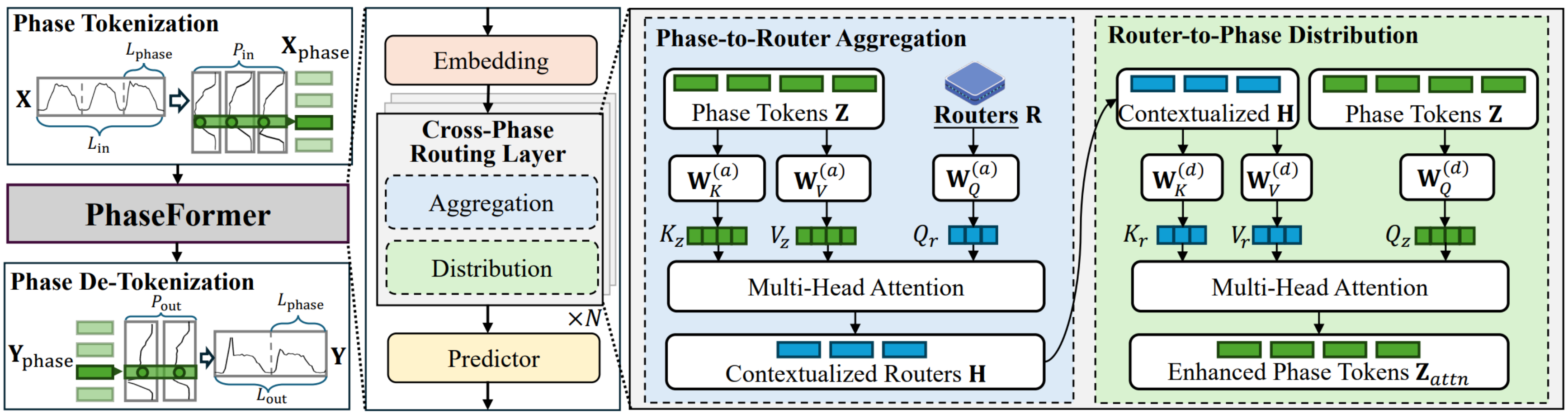
KWANGWOON UNIVERSITY

이번 주 진행사항

- PhaseFormer
  - 아이디어 적용 후 실험
  - 이후 계획

PhaseFormer

Remind



- 입력 시계열 데이터를 주기 단위로 Reshape하고, 2차원 행렬로 차원 변경한 뒤 각 주기의 같은 위상에 있는 값을 묶어 Phase Token으로 사용
- 생성된 Phase Token을 Key와 Value로, Router R을 Query로 사용하여 Multi-Head Attention 수행, 이후 Context 정보가 담긴 Router H를 Key, Value로, Phase Token을 Query로 하여 다시 Multi-Head Attention을 수행해 전역적인 정보가 담긴 Phase Token 생성
- Predictor로 예측값 생성 후 Permute, Reshape 연산을 통해 기존 시계열과 같이 복원

PhaseFormer

실험 세팅

- 사용한 모델: PhaseFormer
- 재현 실험 데이터셋: ETTh1, ETTm1, Weather

Experiment	ETTh1
Learning rate	10^{-3}
Epoch	50
Batch size	256
Loss function	Huber Loss
Seq_len	720
Pred_len	96/192/336/720
Layers	3
Period Len	24
Routers	8~16

PhaseFormer

재현 실험 (ETTh1)

	ETTh1 Reproduction		ETTh1 Reproduction + TCN	
Prediction length	MSE	MAE	MSE	MAE
96	0.364	0.386	**0.349**	**0.383**
192	0.393	0.410	**0.387**	**0.404**
336	0.418	0.424	**0.410**	**0.421**
720	**0.422**	**0.444**	0.431	0.449

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재현 실험 (ETTm1)

	ETTm1 Reproduction		ETTm1 Reproduction + TCN	
Prediction length	MSE	MAE	MSE	MAE
96	**0.299**	**0.345**	0.299	0.346
192	**0.327**	**0.362**	0.342	0.373
336	**0.359**	**0.381**	0.375	0.394
720	**0.417**	**0.414**	0.429	0.424

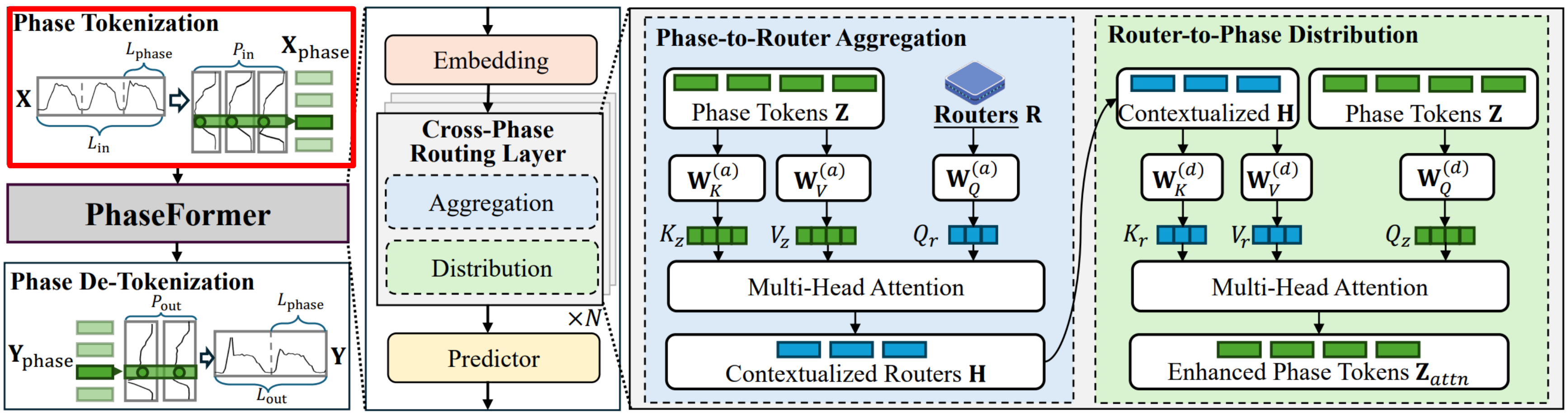
PhaseFormer

재현 실험 (Weather)

	Weather Reproduction		Weather Reproduction + TCN	
Prediction length	MSE	MAE	MSE	MAE
96	**0.148**	**0.193**	0.154	0.197
192	**0.192**	**0.236**	0.196	0.240
336	**0.245**	**0.280**	0.249	0.284
720	**0.320**	**0.335**	0.323	0.335

PhaseFormer

이후 계획



- 논문과 동일하게 FFT를 통해 데이터셋의 주기를 찾을 수 있도록 코드를 수정할 예정

부록

PhaseFormer

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주요 모델 성능 비교

Dataset	Horizon	PhaseFormer		PatchTST		iTransformer		Crossformer		FEDformer		TimeBase		SparseTSF		FITS		TimeMixer		CycleNet		PatchMLP	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.359**	**0.382**	0.377	0.408	0.389	0.421	0.408	0.442	0.485	0.500	0.365	0.387	0.362	0.389	0.380	0.402	0.410	0.441	0.379	0.403	0.417	0.44
	192	**0.397**	**0.404**	0.413	0.431	0.424	0.446	0.472	0.496	0.481	0.498	0.403	0.409	0.404	0.412	0.415	0.424	0.448	0.465	0.416	0.425	0.454	0.465
	336	0.425	0.424	0.436	0.444	0.456	0.469	0.480	0.486	0.522	0.521	**0.409**	**0.419**	0.435	0.426	0.449	0.460	0.475	0.490	0.447	0.445	0.514	0.503
	720	0.431	0.450	0.455	0.475	0.545	0.532	0.710	0.616	0.604	0.575	0.440	**0.448**	**0.426**	0.448	0.433	0.457	0.475	0.500	0.477	0.483	0.548	0.531
ETTh2	96	0.275	0.338	0.276	0.339	0.305	0.361	1.164	0.744	0.401	0.451	0.292	0.350	0.294	0.346	**0.271**	**0.336**	0.315	0.380	**0.271**	0.337	0.324	0.382
	192	0.341	0.376	0.342	0.385	0.405	0.421	1.414	0.830	0.425	0.464	0.339	0.387	0.340	0.377	**0.332**	**0.374**	0.383	0.415	**0.332**	0.38	0.381	0.418
	336	0.369	0.405	0.364	0.405	0.411	0.436	1.220	0.794	0.427	0.471	0.394	0.420	0.360	0.398	**0.355**	**0.396**	0.415	0.436	0.362	0.408	0.408	0.44
	720	0.402	0.436	0.395	0.434	0.448	0.470	2.074	1.103	0.462	0.493	0.400	0.448	0.383	0.425	**0.378**	**0.423**	0.432	0.471	0.415	0.449	0.454	0.469
ETTm1	96	**0.293**	**0.344**	0.298	0.352	0.315	0.369	0.306	0.353	0.406	0.441	0.311	0.351	0.314	0.359	0.313	0.357	0.332	0.384	0.307	0.353	0.319	0.367
	192	**0.323**	**0.361**	0.335	0.373	0.349	0.388	0.341	0.385	0.450	0.477	0.338	0.371	0.348	0.376	0.339	0.369	0.362	0.398	0.337	0.371	0.353	0.385
	336	**0.358**	**0.381**	0.366	0.389	0.381	0.409	0.383	0.420	0.436	0.466	0.364	0.386	0.368	0.386	0.367	0.385	0.386	0.413	0.364	0.387	0.381	0.401
	720	0.412	**0.410**	0.420	0.421	0.437	0.439	0.532	0.512	0.462	0.479	0.413	0.414	0.419	0.413	0.417	0.417	0.452	0.457	**0.41**	0.411	0.436	0.432
ETTm2	96	0.163	0.256	0.165	0.260	0.179	0.274	0.244	0.338	0.339	0.406	0.162	0.256	0.167	0.259	0.166	0.256	0.192	0.285	**0.159**	**0.249**	0.186	0.272
	192	0.219	0.293	0.219	0.298	0.239	0.314	0.350	0.412	0.397	0.452	0.218	0.293	0.219	0.297	0.271	0.328	0.307	0.362	**0.214**	**0.289**	0.25	0.315
	336	0.269	**0.326**	**0.268**	0.333	0.309	0.356	0.400	0.431	0.418	0.452	0.270	0.328	0.271	0.330	0.352	0.380	0.380	0.412	**0.268**	**0.326**	0.298	0.347
	720	**0.351**	**0.379**	0.352	0.386	0.387	0.407	0.574	0.525	0.451	0.499	0.352	0.380	0.353	0.380	0.352	0.380	0.380	0.412	0.353	0.384	0.38	0.398
Weather	96	0.148	**0.195**	0.149	0.199	0.159	0.212	0.151	0.210	0.289	0.342	**0.146**	0.198	0.174	0.231	0.176	0.232	0.163	0.223	0.164	0.220	0.152	0.205
	192	0.193	**0.237**	0.193	0.243	0.203	0.252	0.220	0.273	0.340	0.403	**0.185**	0.241	0.216	0.267	0.203	0.256	0.201	0.255	0.209	0.258	0.197	0.247
	336	0.242	**0.278**	**0.240**	0.281	0.253	0.291	0.287	0.342	0.370	0.408	0.263	0.281	0.260	0.299	0.261	0.299	0.258	0.300	0.255	0.292	0.248	0.287
	720	**0.309**	0.332	0.312	0.334	0.317	0.337	0.362	0.393	0.420	0.421	0.314	**0.331**	0.325	0.345	0.325	0.346	0.329	0.348	0.320	0.338	0.324	0.340
Electricity	96	0.129	**0.221**	0.141	0.240	0.135	0.233	0.140	0.237	0.226	0.341	0.139	0.231	0.139	0.239	0.147	0.253	0.142	0.247	**0.128**	0.223	0.141	0.243
	192	0.148	0.238	0.156	0.256	0.155	0.253	0.165	0.259	0.220	0.336	0.153	0.245	0.155	0.250	0.159	0.256	0.164	0.273	**0.143**	**0.237**	0.163	0.265
	336	0.165	0.257	0.172	0.267	0.169	0.267	0.190	0.286	0.224	0.337	0.169	0.262	0.171	0.265	0.169	0.270	0.171	0.260	**0.159**	**0.254**	0.186	0.290
	720	0.201	**0.285**	0.208	0.299	0.204	0.301	0.227	0.312	0.271	0.378	0.207	0.294	0.208	0.300	0.214	0.302	0.209	0.313	**0.197**	0.287	0.221	0.320
Traffic	96	**0.361**	**0.238**	0.363	0.250	0.374	0.273	0.512	0.265	0.664	0.431	0.394	0.267	0.389	0.272	0.374	0.273	0.404	0.293	0.381	0.266	0.402	0.302
	192	**0.373**	**0.243**	0.382	0.258	0.393	0.283	0.528	0.271	0.613	0.382	0.407	0.270	0.399	0.272	0.393	0.282	0.404	0.292	0.394	0.273	0.419	0.309
	336	**0.385**	**0.248**	0.399	0.268	0.409	0.292	0.543	0.281	0.612	0.379	0.417	0.278	0.417	0.279	0.423	0.292	0.425	0.293	0.406	0.279	0.437	0.317
	720	**0.428**	**0.270**	0.432	0.289	0.450	0.314	0.598	0.314	0.664	0.410	0.456	0.298	0.449	0.299	0.450	0.314	0.453	0.314	0.441	0.300	0.479	0.342

PhaseFormer

Ablation Study

Table 2: Cross-Phase Routing layer ablation. Each cell reports MSE, MAE, and FLOPs. Lower is better for all metrics. FLOPs are reported in millions (MFLOPs). The best results are highlighted with **Bold**, and the second-best results with Underlined.

Dataset	PhaseFormer			w/ FullAttention			w/ LinearMixing			w/o Routing		
	MSE	MAE	FLOPs	MSE	MAE	FLOPs	MSE	MAE	FLOPs	MSE	MAE	FLOPs
Weather	**0.1503**	**0.1971**	3.119	0.1527	0.2005	3.202	0.1700	0.2226	0.920	0.1907	0.2406	**0.783**
Electricity	**0.1290**	**0.2209**	42.213	0.1295	0.2217	48.951	0.1403	0.2334	14.068	0.1423	0.2365	**11.972**
Traffic	**0.3721**	**0.2475**	113.356	0.3791	0.2513	131.452	0.3842	0.2532	37.776	0.3892	0.2584	**32.149**

- 기존 Full Attention 방식, LinearMixing, Routing 메커니즘 제거 등과 같이 실험해본 결과, PhaseFormer의 성능이 가장 좋은 것을 확인할 수 있었음

PhaseFormer

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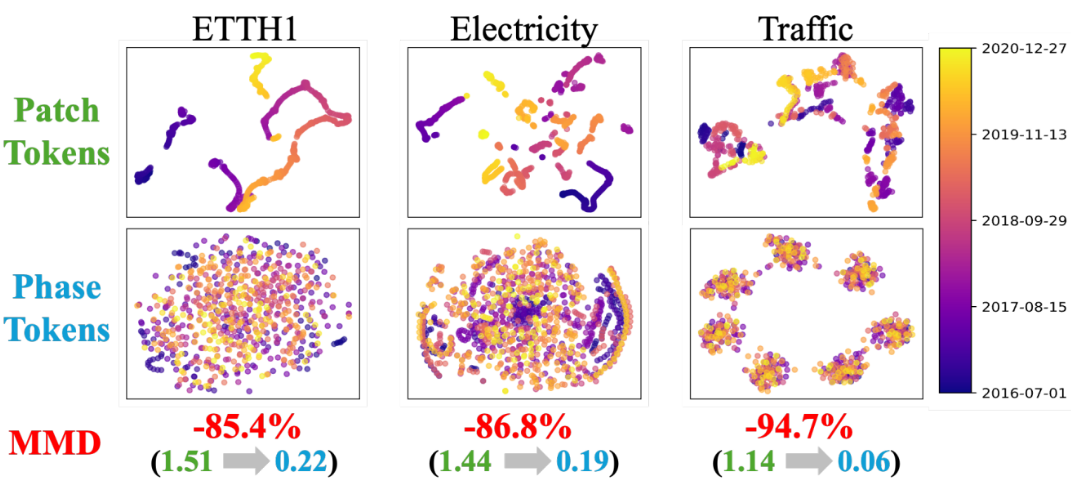
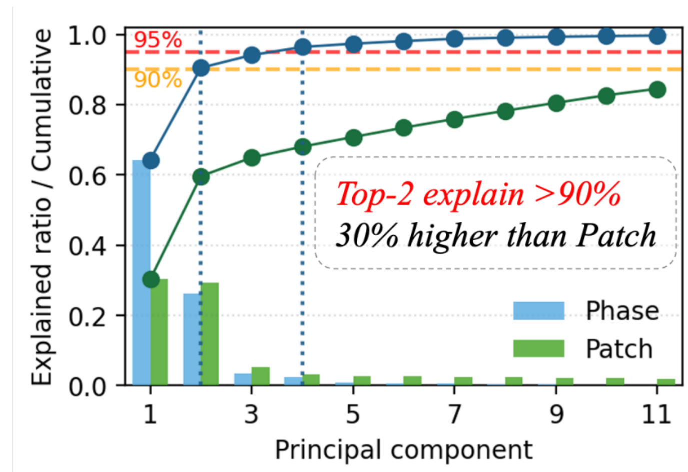
Motivations



(a) Temporal shift comparison across multiple datasets.



(b) PCA for phase tokens on *Traffic*.

- (a)의 경우, 기존 Patch Tokens은 시간이 지남에 따라 분포가 이동하여 예측이 어렵지만, Phase Tokens은 안정적인 예측이 가능해짐
- (b)의 경우, PCA(주성분 분석) 결과, Phase Tokens은 상위 2개의 차원만으로 전체 분산의 90% 이상을 설명 가능함

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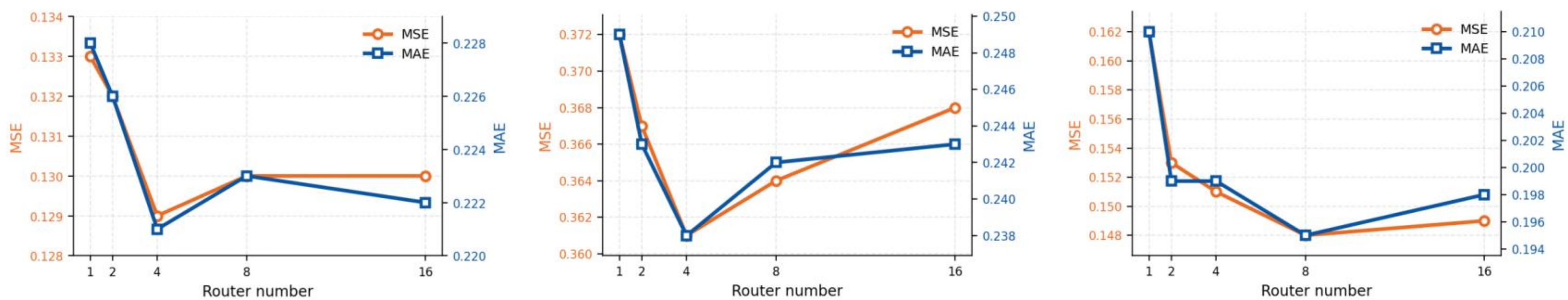
Varying the Number of Routers



(a) Electricity (b) Traffic (c) Weather

Figure 5: Effect of varying the number of routers M on forecasting performance on three datasets.

- Router의 개수별 성능 비교, Router 수가 증가함에 따라 예측 오차가 감소하다가 약간 상승함
- 대부분 4~8개의 성능이 가장 좋은 것을 확인